UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

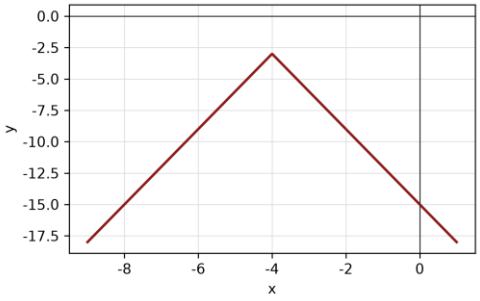
INSTRUCTIONS TO MARKERS

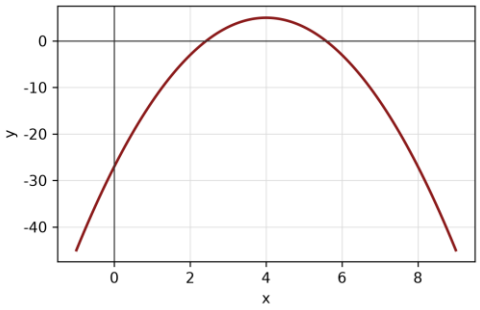
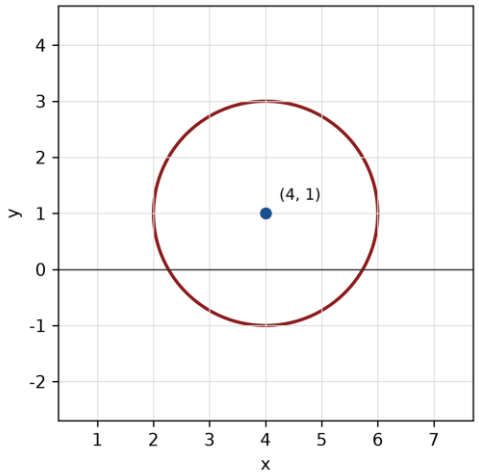
1. This marking scheme accompanies the MAT133 mock paper (October 2026).
2. Answers and marking points are for local study use only.

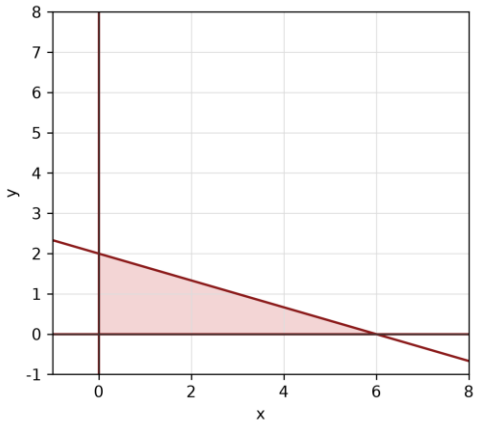
FOR MARKERS' USE ONLY

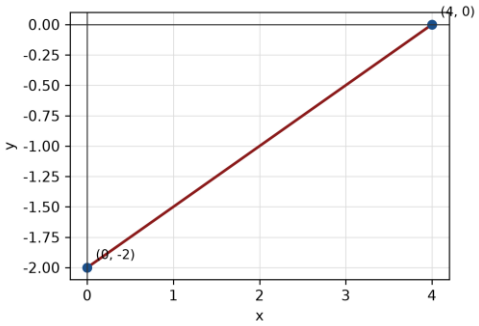
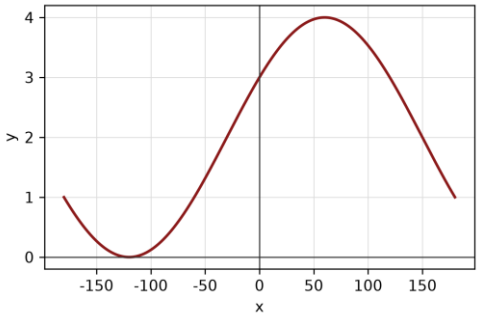
This marking scheme consists of 6 printed pages

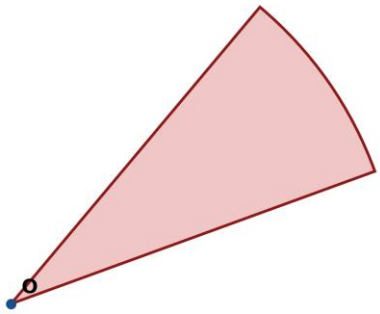
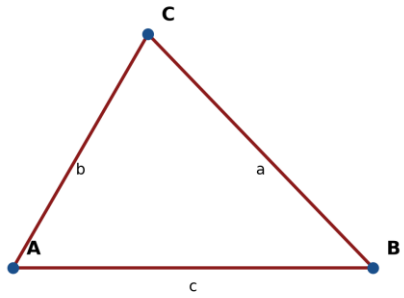
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NO	SOLUTION		MARK
1 a)	$-3 < 3x - 1 < 2$	M1	4 m
	$-2 < 3x < 3$	M1	
	$-\frac{2}{3} < x < 1$	A1	
			
1 b)	$\|5x - 2\| - 7 > 0$	M1	5 m
	$2 - 5x > 7$ or $2 - 5x <= -7$	A1	
	$(-\infty, -1) \cup \left(\frac{9}{5}, \infty\right)$	A1	
			
1 c)	$y = -3\|x + 4\| - 3$	B1	5 m
	vertex $(-4, -3)$	A1	
	no real intercepts	A1	
			
2 a)	$-1x + 4 \neq 0$	M1	4 m
	$x \neq 4$	A1	
	$3x + -5 \geq 0$	M1	
	$x \geq \frac{5}{3}$	A1	

NO	SOLUTION		MARK
2 b)	$y = 5 - 2(x-4)^2$ vertex $(4, 5)$ $y(0) = -27$ 	B1 B1 A1 A1	5 m
2 c)	$(x-4)^2 + (y-1)^2 = 4$ $x^2 - 8x + y^2 - 2y + 17 = 4$ $x^2 - 8x + y^2 - 2y + 13 = 0$ 	M1 A1 B1	5 m
3 a)	$\frac{2+4i}{3+2i}$ $= \frac{(2+4i)(3-2i)}{(3+2i)(3-2i)}$ $= \frac{14+8i}{13}$ $\frac{14}{13} + \frac{8i}{13}$	M1 M1 A1 A1	5 m

NO	SOLUTION		MARK
3 b)	$\lvert z\rvert = \sqrt{-5^2 + 12^2} = 13$ $\tan\theta = \frac{12}{-5}, \quad \theta = 112.6^\circ$	M1 A1 M1 A1	5 m
3 c)	$x^3 - 2x^2 - 9x + 18$ $f(3) = 0, \quad f(-3) = 0, \quad f(2) = 0$ $(x-3)(x-2)(x+3)$	M1 M1 A1	5 m
4 a)	$-3x + 3y + z = -14$ $2x + 2y - 3z = 2$ $2x + 3y + 3z = -13$ $x = 1, \quad y = -3, \quad z = -2$	M1 A1 A1	5 m
4 b)	$x = 0, \quad y = 0, \quad 1x + 3y = 6$ $x \geq 0, \quad y \geq 0, \quad 1x + 3y \leq 6$  	B1 B1 A1	5 m

NO	SOLUTION	MARK
4 c)	$\frac{x}{4} - \frac{y}{2} = 1$ M1 $-2x + 4y = -8$ $y = \frac{x}{2} - 2$ A1 	4 m
5 a)	amplitude $= 2$ A1 A1 period $= \frac{360^\circ}{1} = 360^\circ$ A1 phase shift $= -\frac{30}{1} = -30^\circ$ B1 	5 m

NO	SOLUTION	MARK
5 b)	$\theta = 30^\circ \times \frac{\pi}{180} = \frac{\pi}{6}$ M1 M1 $s = r\theta = 6\left(\frac{\pi}{6}\right) = \pi$ A1 	4 m
5 c)	$a^2 = b^2 + c^2 - 2bc\cos A$ M1 $a^2 = 9^2 + 12^2 - 2(9)(12)\cos 120^\circ = 333$ A1 $a = 3\sqrt{37}$ A1 	4 m